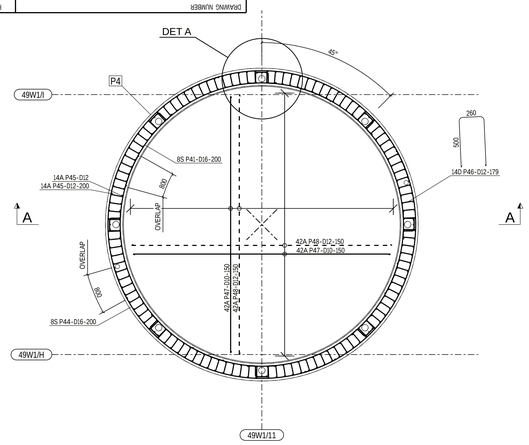
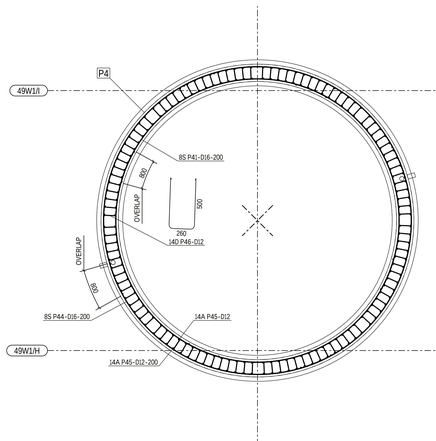
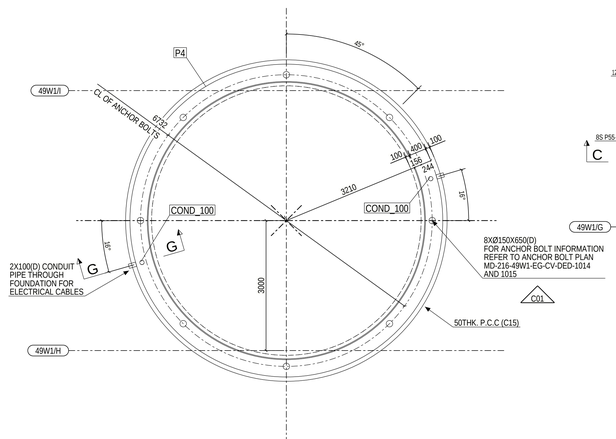


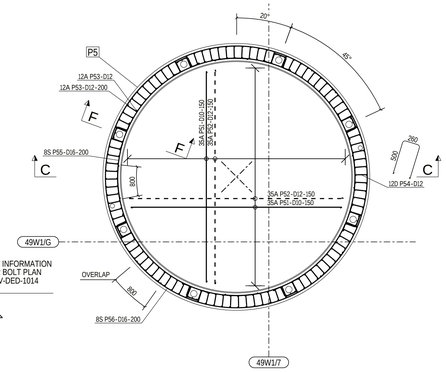
49W1-TNK-003 R.C.C DETAIL AT TOP OF FOUNDATION
(SCALE 1:50)



49W1-TNK-003 R.C.C DETAIL AT BOTTOM OF FOUNDATION
(SCALE 1:50)



49W1-TNK-003 PLAN AT TOP OF FOUNDATION
(SCALE 1:50)



49W1-TNK-009 R.C.C DETAIL AT TOP OF FOUNDATION
(SCALE 1:50)

CONCRETE INFORMATION

Concretes used	C35
Concrete cover 1	50±10mm

REINFORCING BAR LIST

CAST UNIT CODE: P4 — PCS: 1

REINFORCING BARS TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
S	P41	32	B500B	16	5930		300.1	5937	3276						***	***	
R	P42	56	B500B	12	1110		55.5	200	265	305	265	200					60
A	P43	24	B500B	12	680		14.6	685									
S	P44	32	B500B	16	6370		322.4	6379	3560						***	***	
A	P45	224	B500B	12	1270		253.6	1275									
D	P46	224	B500B	12	1200		238.9	500	260	500							60
A	P47	84	B500B	10	6150		248.1	6153									
16 AT GROUP	42	PCS		...	1480	114	...	1489									
A	P48	84	B500B	12	6150		357.0	6153									
2 AT GROUP	42	PCS		...	1490	114	...	1495									

REINFORCING BAR TOTAL WEIGHT [kg] 1790.1

REINFORCING BAR LIST

CAST UNIT CODE: P5 — PCS: 1

REINFORCING BARS TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
A	P51	70	B500B	10	5120		171.2	5120									
2 AT GROUP	35	PCS		...	1300	112	...	1306									
A	P52	70	B500B	12	5120		246.4	5120									
2 AT GROUP	35	PCS		...	1310	112	...	1312									
A	P53	192	B500B	12	1270		217.4	1275									
D	P54	192	B500B	12	1200		204.7	500	260	500							60
S	P55	32	B500B	16	5100		258.0	5105	2748						***	***	
S	P56	32	B500B	16	5540		280.3	5546	3030						***	***	
R	P57	48	B500B	12	1110		47.5	200	265	305	265	200					60
A	P58	24	B500B	12	560		11.9	560									

REINFORCING BAR TOTAL WEIGHT [kg] 1437.4

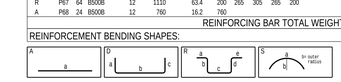
REINFORCING BAR LIST

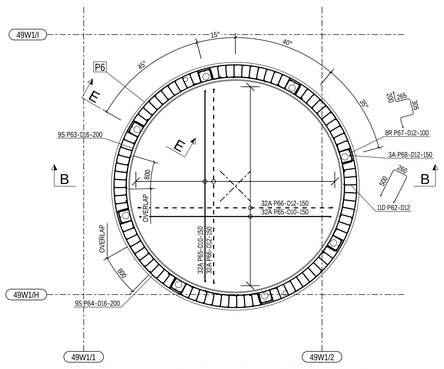
CAST UNIT CODE: P6 — PCS: 1

REINFORCING BARS TYPE	POS	PCS	GRADE	D [mm]	L [mm]	dL [mm]	WEIGHT SUM [kg]	BENDING DIMENSIONS [mm] a	b	c	d	e	u	v	x	TD	COMMENT
A	P61	176	B500B	12	1470		230.5	1475									
D	P62	176	B500B	12	1200		187.7	500	260	500							60
S	P63	36	B500B	16	4750		270.6	4759	2526						***	***	
S	P64	36	B500B	16	5190		295.7	5201	2810						***	***	
A	P65	64	B500B	10	4650		143.2	4653									
31 AT GROUP	32	PCS		...	1220	111	...	1228									
A	P66	64	B500B	12	4650		206.1	4653									
2 AT GROUP	32	PCS		...	1230	110	...	1234									
R	P67	64	B500B	12	1110		63.4	200	265	305	265	200					60
A	P68	24	B500B	12	760		16.2	760									

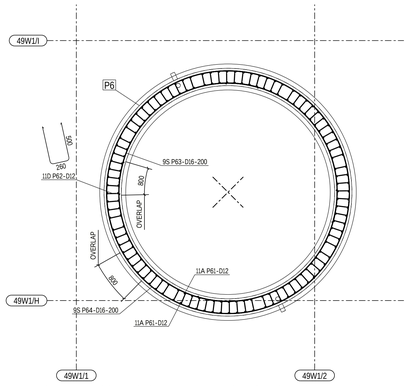
REINFORCING BAR TOTAL WEIGHT [kg] 1413.4

REINFORCEMENT BENDING SHAPES:

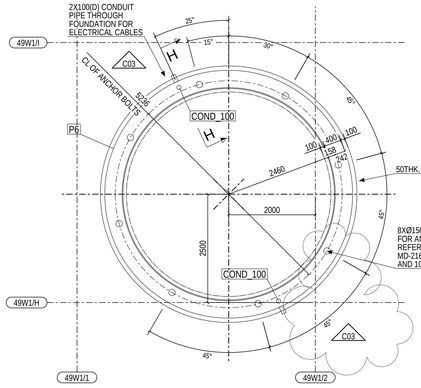




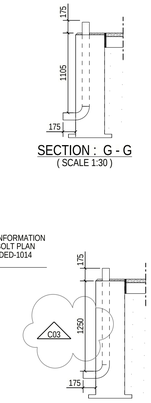
49W1-TNK-007 R.C.C DETAIL AT TOP OF FOUNDATION
(SCALE 1:50)



49W1-TNK-007 R.C.C DETAIL AT BOTTOM OF FOUNDATION
(SCALE 1:50)

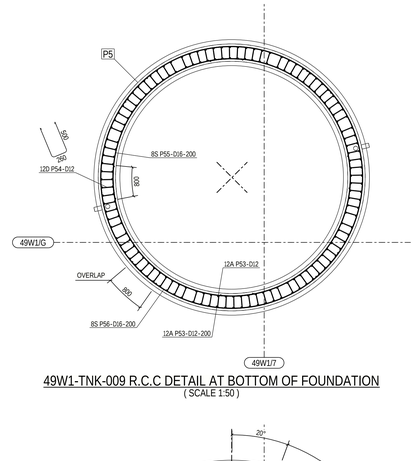


49W1-TNK-007 PLAN AT TOP OF FOUNDATION
(SCALE 1:50)

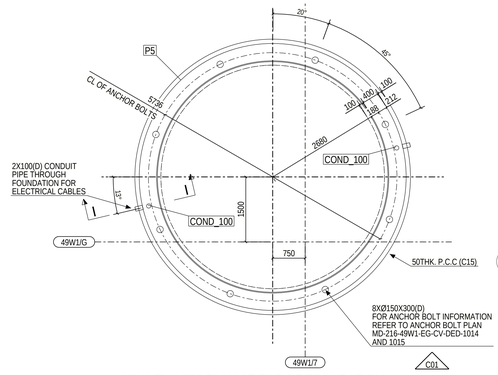


SECTION : G - G
(SCALE 1:30)

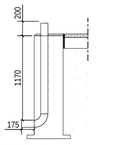
SECTION : H - H
(SCALE 1:30)



49W1-TNK-009 R.C.C DETAIL AT BOTTOM OF FOUNDATION
(SCALE 1:50)



49W1-TNK-009 PLAN AT TOP OF FOUNDATION
(SCALE 1:50)



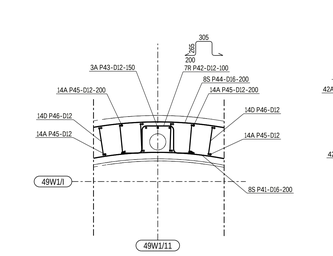
SECTION : I - I
(SCALE 1:30)

LEGEND:

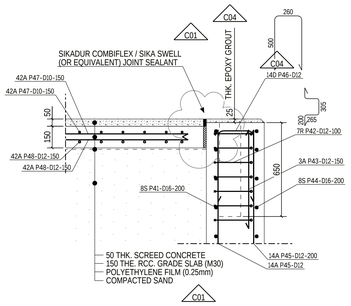
R.L - REDUCED LEVEL
T.O.C. - TOP OF CONCRETE
B.O.F. - BOTTOM OF FOUNDATION
F.G.L. - FINISHED GROUND LEVEL
P.C.C. - PLAIN CEMENT CONCRETE
TYP - TYPICAL
MKD - MARKED
CL - CENTER LINE
BOTTOM REINF. (B)
TOP REINF. (T)
SBC - SAFE BEARING CAPACITY
MIN. - MINIMUM
REF. - REFERED
GA - GENERAL ARRANGEMENT
DWG. - DRAWING
R.C.C. - REINFORCED CEMENT CONCRETE

NOTES:

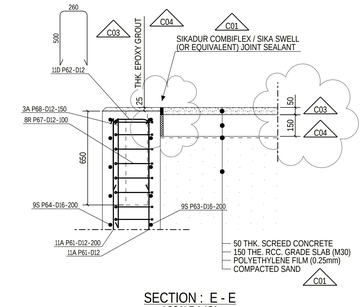
1. ALL DIMENSIONS ARE IN MILLIMETRES AND LEVELS ARE IN METRES UNLESS OTHERWISE STATED
2. GRADE OF CONCRETE SHALL BE C35 (35MPa) UNLESS NOTED OTHERWISE & C15 (15MPa) FOR BUNDING CONCRETE.
3. GRADE OF REINFORCEMENT SHALL BE DEFORMED BAR ASTM A615-GRADE 60 (fy=420MPa)
4. CLEAR COVER TO MAIN REINFORCEMENT SHALL BE AS FOLLOWS:
   a) FOUNDATION: 75 mm (BOTTOM & SIDE)
   b) FOUNDATION: 50 mm (TOP)
   c) COLUMN & PEDESTAL: 50 mm
5. MINIMUM LAP LENGTHS OF REINFORCEMENT SHALL BE 40XBAR DIAMETER
6. FOR STANDARD REINFORCEMENT DETAILS & GENERAL NOTES FOR CIVIL WORKS REFER TO DRAWING NUMBER MD-216-1000-EG-CV-DED-1005 & 1001
7. ANCHOR BOLTS AND EMBEDDED STEELWORK SHALL BE HOT DIP GALVANIZED
8. T.O.C. & TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10MM OF DESIGN LEVEL
9. NET SAFE BEARING CAPACITY OF SOIL AT BOTTOM OF FOUNDATION @ NET SAFE BEARING CAPACITY UNDER RING WALL 375 kPa
10. BARS CAN BE MOVED LOCALLY (+/- 75mm) TO AVOID INTERFERENCE WITH EMBEDDED PARTS C01
11. SAND SHALL BE COMPACTED TO A MAXIMUM DEPTH OF 300mm & SHALL BE COMPACTED TO 80% RELATIVE DENSITY AS DETERMINED BY ASTM D4253 & ASTM D254
12. TOP OF SCREED TOLERANCE SHALL BE WITHIN +/- 10mm OF DESIGN LEVEL



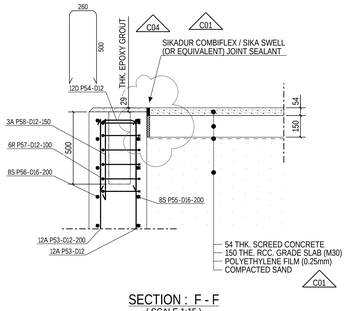
DETAIL : A
(SCALE 1:20)



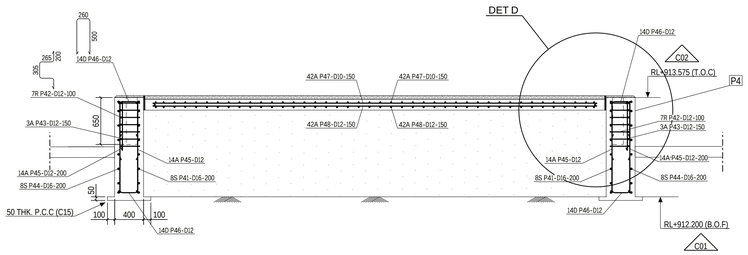
DETAIL : D
(SCALE 1:15)



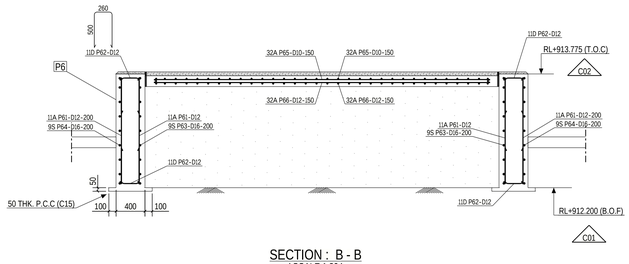
SECTION : E - E
(SCALE 1:15)



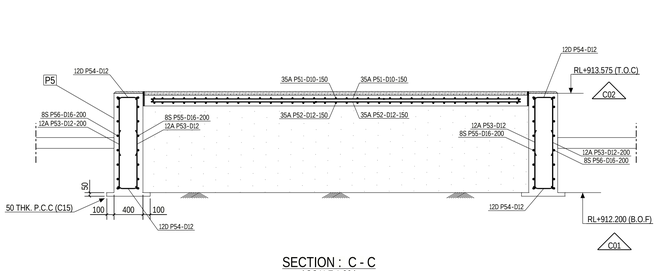
SECTION : F - F
(SCALE 1:15)



SECTION : A - A
(SCALE 1:30)

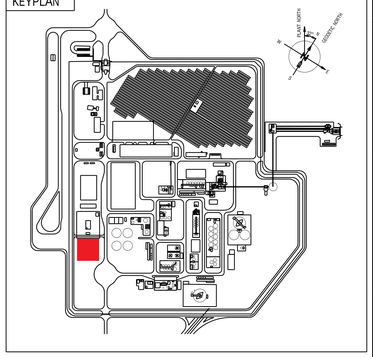


SECTION : B - B
(SCALE 1:30)



SECTION : C - C
(SCALE 1:30)

KEYPLAN



REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
D01	29.01.2024	AS-BUILT	JonRen	MauNot	NanWu	LuiPiz	
C04	14.01.2022	49W1-TNK-007 AND 009 GROUT THICKNESS CHANGED, GROUT --> EPOXY GROUT	JonRen	MauNot	NanWu	LuiPiz	
C03	29.12.2021	49W1-TNK-007 BOLT- AND CONDUIT POSITION CHANGED, GROUT THICKNESS INCREASED	JonRen	MauNot	NanWu	LuiPiz	
C02	17.12.2021	FOUNDATION TOP LEVEL CHANGED	JonRen	MauNot	NanWu	LuiPiz	
C01	19.11.2021	ISSUED FOR CONSTRUCTION, REVISED AS FROM COMMENTS	JonRen	MauNot	NanWu	LuiPiz	
A01	29.10.2021	ISSUED FOR APPROVAL	JonRen	MauNot	NanWu	LuiPiz	

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-CV-CAL-1001	TANK FOUNDATIONS CALCULATION REPORT
MD-216-49W1-EG-CV-GAD-1003	1-2/1/A-E GENERAL ARRANGEMENT DRAWING
MD-216-49W1-EG-CV-GAD-1001	1-8/1/E-J GENERAL ARRANGEMENT DRAWING

DOCUMENT NUMBER	REFERENCE DOCUMENTS
MD-216-49W1-EG-CV-DED-1001	BUFFER TANK FOUNDATIONS (REA015 AND REA003) DETAIL- AND REINFORCEMENT DRAWING
MD-216-49W1-EG-CV-GAD-1002	7-12/D-J GENERAL ARRANGEMENT DRAWING

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

LARSEN & TOUBRO

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LARSEN & TOUBRO DRAWING NUMBER	0190445_080_C_D_24892_002
REVISION	5
OUTOTEC DRAWING NUMBER	900707_GAU01_AG11_CTB01_49002
DOCUMENT ID	OU100732634
REVISION	5

DRAWN	JonRen	29.10.2021
CHECKED	MauNot	29.10.2021
ENGINEER	NanWu	29.10.2021
APPROVED	PiuPiz	29.10.2021
EMPLOYER		
SCALE	1:50	A0

MANSOURAH - MASSARAH GOLD PROJECT
49W1 RAW WATER TREATMENT PLANT
TANK FOUNDATIONS 49W1-TNK-003/007/009
DETAIL- AND REINFORCEMENT DRAWING

DRAWING NUMBER: MD-216-49W1-EG-CV-DED-1002 | REV: D01